Intermediate Geometry
BAPS-BUBT National Camp 2025

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Vector Geometry

Dot Product

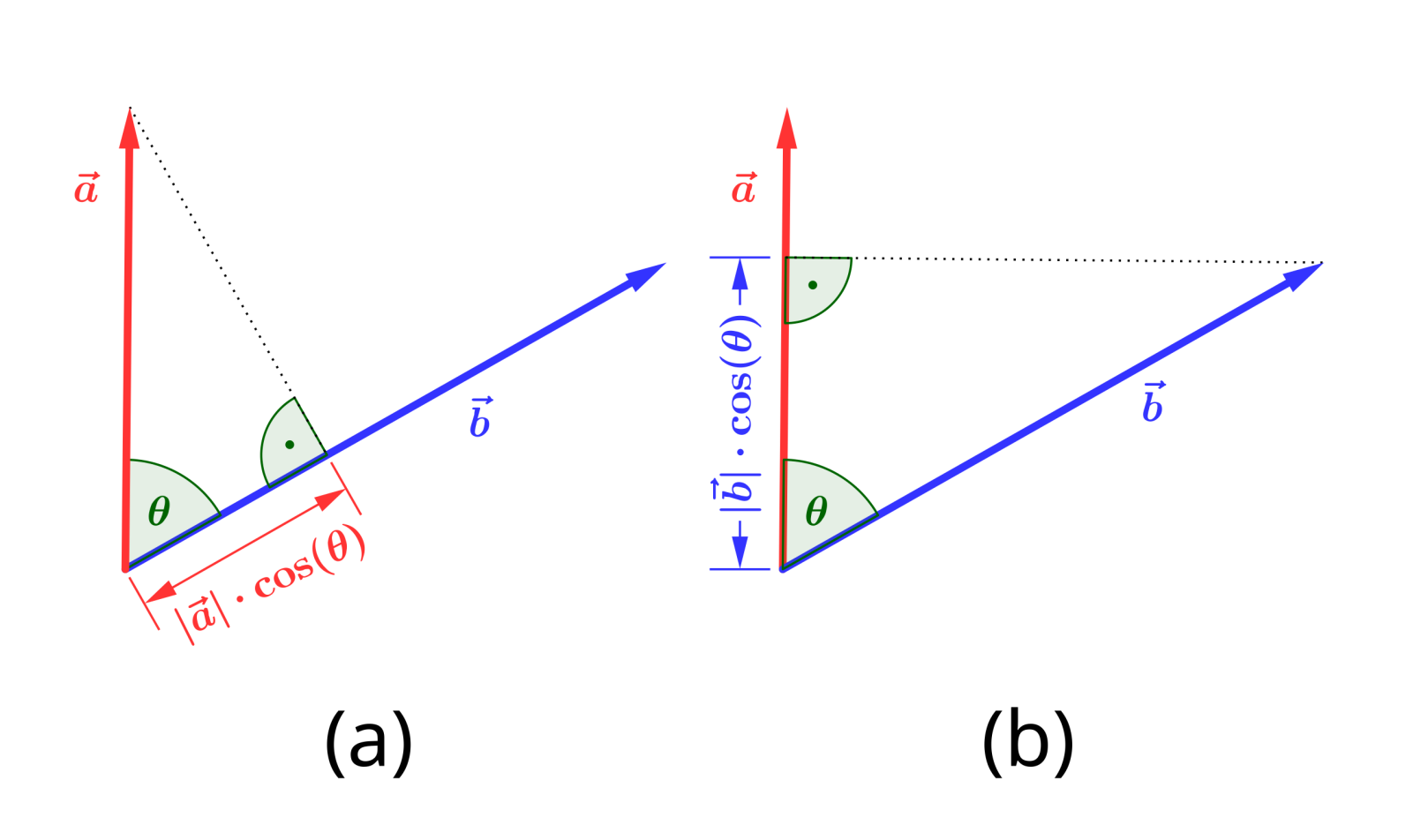


Figure 1: Wikimedia Commons, Dot Product

Dot Product

- Given two vectors $\mathbf{a}$ and $\mathbf{b}$, their dot product is given by two equivalent definitions:

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos\theta$$

  where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$.
- In 2D, we also have $\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y$.
- In 3D, we have $\mathbf{x} \cdot \mathbf{y} = a_x b_x + a_y b_y + a_z b_z$.
- It is usually used to find the projection of a vector on another vector.
- Also, two perpendicular vectors have dot product equal to 0.

Line-Point Distance

- Let $\mathbf{r}$ be a unit vector along the line.
- We can compute it by setting $\mathbf{r} = (B - A)/\|B - A\|$, where B and A are two distinct points on the line.
- Let $\mathbf{n} = (r_y, -r_x)$ be a unit normal to the line.
- Then the distance d between the line and a point P is given by $|n \cdot B - n \cdot P|$.

Cross Product

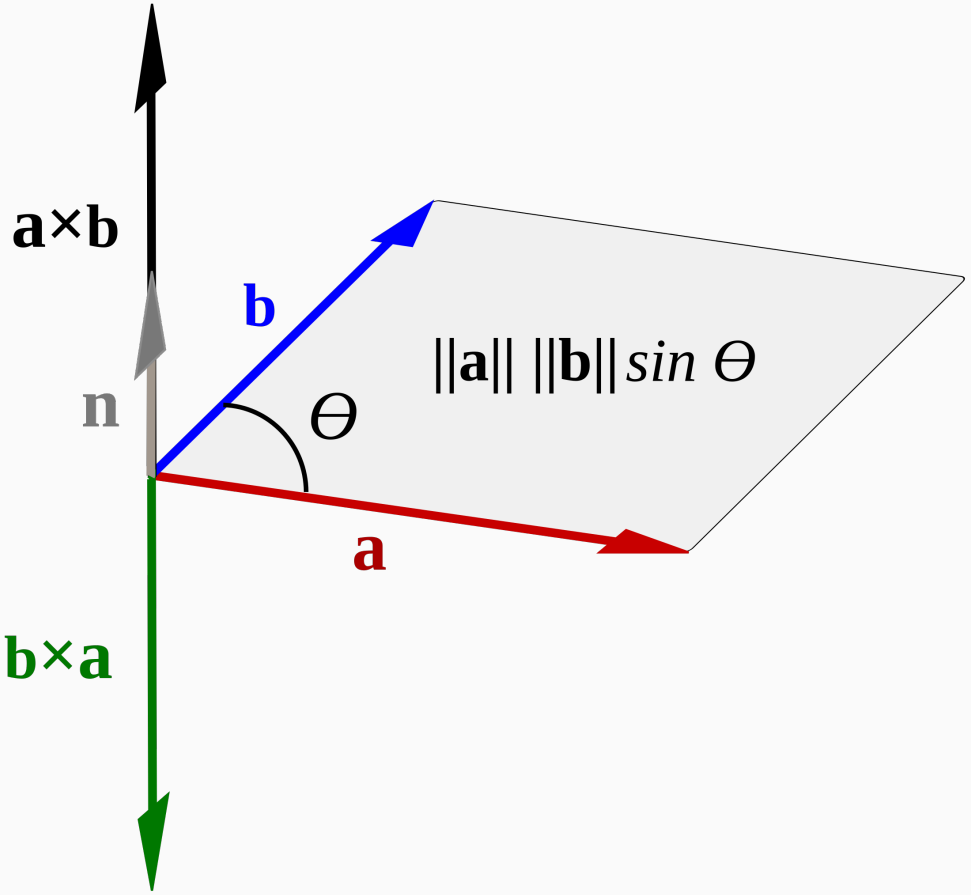


Figure 2: Wikimedia Commons, Cross Product

Cross Product

- The cross product is a 3D operation
- $\mathbf{a} \times \mathbf{b} = \mathbf{n} \cdot \|\mathbf{a}\| \|\mathbf{b}\| \sin \Theta$
- $\mathbf{n}$ is a unit normal vector to the plane defined by $\mathbf{a}$ and $\mathbf{b}$.
- We often use its '2D-version', where we simply set the z-coordinates to 0.
- In this case, the cross-product result is of the form $(0, 0, z)$, where $z > 0$ if $\mathbf{b}$ is counter-clockwise to $\mathbf{a}$.
- If $z = 0$, then the two lines are parallel.

Ternary Search

Problem Statement

You are given a circle representing the Moon and two other points representing the location of Alice and Bob respectively. Find the length of the shortest path from Alice to Bob that also touches the Moon at least once.

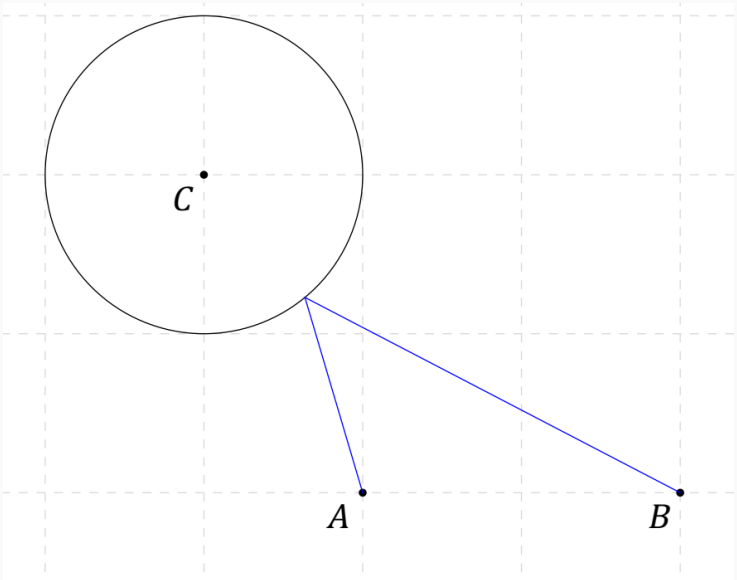


Figure 3: Sample Input along with Optimal Path

CERC 2023, Going to the Moon

- A lot of geometry problems can be solved using ternary search and a little case-work.
- There are a few cases here.
- The direct line segment between Alice and Bob coud already be optimal.
- Or the optimal path touches the Moon at exactly one point.
- We can ternary search for this point on some arc containing this optimal point.
- It is important to be sure (or sort-of sure) that the function we're optimising using ternary search is convex/concave in our search space.
- For geometry problems, it is often enough to visualize the case to convince yourself.

Sweeping

Closest Pair of Points

Problem Statement

You are given n pair of points ($1 \leq n \leq 10^5$), the ith of which has co-ordinates (x_i, y_i), where $|x_i|, |y_i| \leq 10^9$ and x_i and y_i are integers. Find the distance between the closest pair of points.

Closest Pair of Points

- Line sweep!
- More precisely, we handle the points in ascending order of their x co-ordinates.
- We maintain h, the smallest distance found so far.
- Then the next point (x, y) to be inserted can update h only if there is $(x', y') \in [x - h, x] \times [y - h, y + h]$.
- It can be shown that there are at most constantly many such (x', y') at any step.

Area of the Union of Rectangles

Problem Statement

We are given n ($1 \leq n \leq 10^5$) rectangles whose sides are parallel to the x or the y-axis. The co-ordinates of the corners of the rectangles are at most 10^9 in their absolute values. Find the total area of the union of the given rectangles.

Area of the Union of Rectangles

- Compress the y-coordinates.
- Take a vertical line and sweep it left to right.
- Use segment tree to maintain set of active vertical 'segments'.
- Multiply by change in x-coordinate to get updated area.

Problem Statement

There are n islands ($1 \le n \le 2 \cdot 10^5$), the ith of which is represented by co-ordinates (x_i, y_i) ($|x_i|, |y_i| \le 10^9$). We can place a radar at $(0, 0)$. If the radar has a scanning angle range of α and we rotate it to an angle of θ, then it can monitor all islands located within the angular range $\left[\theta - \frac{\alpha}{2}, \theta + \frac{\alpha}{2}\right]$.

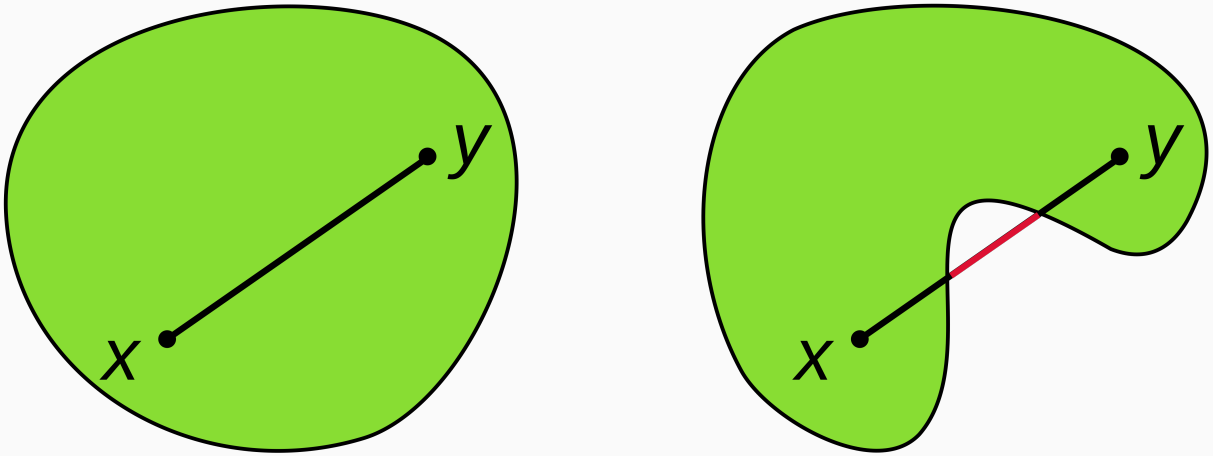
Given k ($1 \le k \le n$), find the minimum $\alpha \ge 0$ such that for any θ, the radar can monitor at least k islands.

- This problem is actually quite easy
- Radial sweep! Sort the islands counter-clockwise
- Select the maximum angle difference between the ith and the $i + k$th island (modulo n).
- There may be a few details that I have not mentioned in this slide.

Convex Hull

Convex Set

A set of points S in the 2D plane is called convex if and only if for every pair of points in S, the line segment connecting them lies entirely in S.



(a) Convex (b) Non-convex

Figure 4: Convex and non-convex, Wikimedia Commons

Convex Hull

- The convex hull of a set S of points is the smallest convex set that contains S.
- When S is a finite set of points, the convex hull is a polygon: a convex polygon.

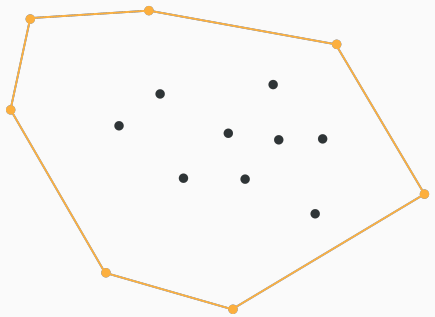


Figure 5: Convex Hull, Wikimedia Commons

Graham's Scan

- This algorithm finds the convex hull of n points in the 2D plane in $O(n \log n)$ time.
- This is essentially a radial sweep algorithm.
- We first pick the bottom-most and then left-most point P_0.
- We sort the other points in counter-clockwise order by their angle relative to P_0 (if the angles are same, then they are sorted in ascending order of their distance from P_0).
- Then we sweep through in the sorted order, incrementally building the convex hull.
- As we add a point, we utilise the fact that a convex polygon has no internal angle over 180°.
- While the previous two points and the current point together make a clockwise turn, we discard the immediate previous point.

Problem Statement

You are given n ($4 \leq n \leq 4000$) distinct points $p_1, \ldots, p_n$ on a 2D plane, such that no three points are collinear. Find four distinct integers i, j, k, and l such that p_i lies strictly inside the triangle formed by points p_j, p_k, and p_l. If no such exist, output -1. Also, the co-ordinates of p_i is given by (x_i, y_i), where x_i and y_i are integer points such that $1 \leq x_i, y_i \leq 5 \cdot 10^5$.

- Let C denote the points belonging to the convex hull of $p_1, \ldots, p_n$.
- What happens when all the points p_i belong to C?
- Suppose that there is some i such that p_i does not belong to C. Can we do anything here?

Resources/References

- CP-Algorithm Convex Hull
- USACO Guide Sweep Line
- CodeForces Sweep Line
- TopCoder